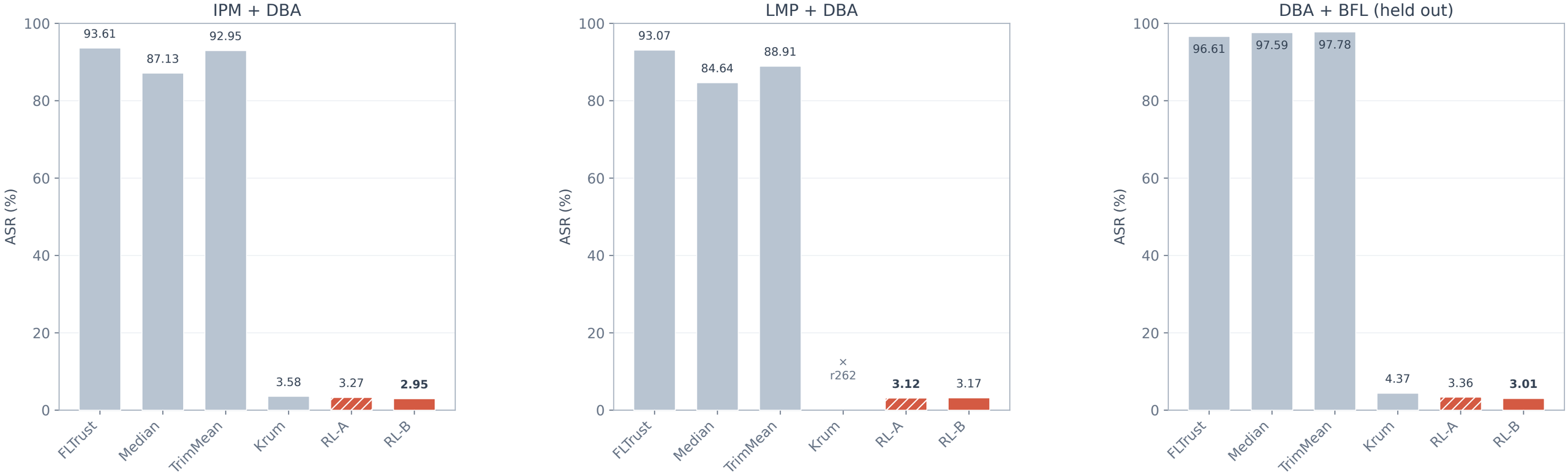
Shared RL under mixed attacks: ASR

CIFAR-10 / CIFAR10CNN / IID / 20 clients, all participate / 4 malicious / frozen 500-round queries



RL-A: pooled single-attack trajectories RL-B: simultaneous-pair trajectories Conventional defenses

Bars: query rounds 491–500 means; labels: 2 decimals. Bold = best observed value per scenario. One seed; no error bars.

Local / root steps 5; malicious steps 10; batch 128; SGD lr 0.05; train / query seed 42 / 43. Two attack types use fixed 2 + 2 groups.

× = attack construction failed; no endpoint bar. DBA+BFL / BFL held out from training. Both RL variants use Histogram40 and 10k cached training steps.